

Experiment 15: Git Branching and Merge Conflict Resolution

Aim

To demonstrate the use of Git branches and resolve merge conflicts while collaborating on a project using Git and GitHub.

Tools Required

- Git
- GitHub
- VS Code (or any text editor)

Repository Used

Repository: <https://github.com/Sriarevalagan23/Test-repo-fork/tree/main>

Procedure

Step 1: Clone the Repository

Bash

```
git clone https://github.com/Sriarevalagan23/Test-repo-fork/tree/main
cd repository
```

Screenshot 1: Terminal showing successful clone.

A screenshot of a terminal window titled "Desktop — -zsh — 80x24". The terminal shows the following commands and output:

```
Last login: Tue Jul 21 22:18:52 on ttys000
[cd desktop]
srinidhi@Srinidhis-Laptop ~ % cd desktop
[srinidhi@Srinidhis-Laptop desktop % git clone https://github.com/Sriarevalagan23/Test-repo-fork
Cloning into 'Test-repo-fork'...
remote: Enumerating objects: 70, done.
remote: Counting objects: 100% (70/70), done.
remote: Compressing objects: 100% (63/63), done.
remote: Total 70 (delta 1), reused 70 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (70/70), 1.41 MiB | 6.25 MiB/s, done.
Resolving deltas: 100% (1/1), done.
srinidhi@Srinidhis-Laptop desktop %
```

Step 2: Create and Switch to a New Branch

Bash

```
git checkout -b feature-update
```

Screenshot 2: Branch creation in terminal.

```
[srinidhi@Srinidhis-Laptop Test-repo-fork % git branch
main
* update-readme
srinidhi@Srinidhis-Laptop Test-repo-fork % git checkout -b feature-update

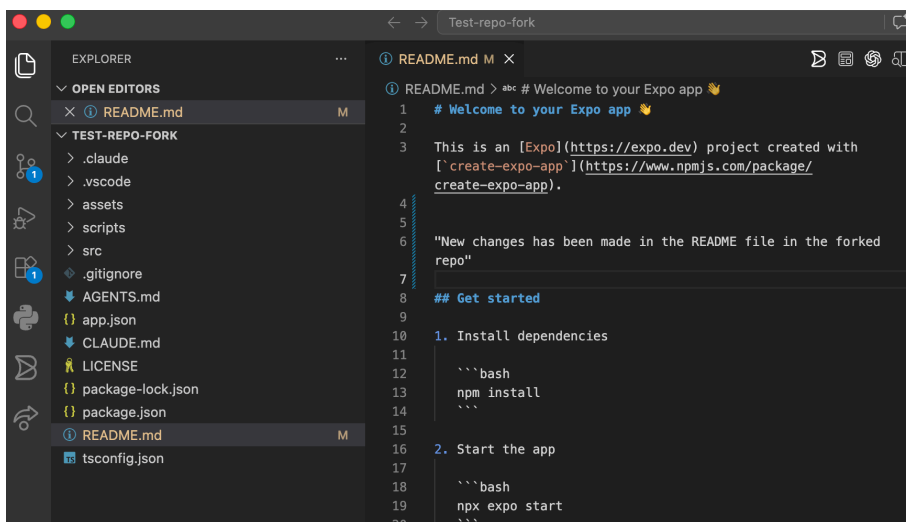
Switched to a new branch 'feature-update'
srinidhi@Srinidhis-Laptop Test-repo-fork %
```

Step 3: Modify a File

Example:

- Update the README.md file.
- Add a new description or modify an existing line.

Screenshot 3: Modified file in VS Code.



Step 4: Commit the Changes

```
Bash
git add .
git commit -m "Updated README in feature branch"
```

Screenshot 4: Commit success.

Step 5: Push the Branch

```
Bash
git push origin feature-update
```

Screenshot 5: Push completed.

```

srinidhi@Srinidhis-Laptop Test-repo-fork % git push origin feature-update

Enumerating objects: 76, done.
Counting objects: 100% (76/76), done.
Delta compression using up to 8 threads
Compressing objects: 100% (69/69), done.
Writing objects: 100% (76/76), 1.41 MiB | 2.21 MiB/s, done.
Total 76 (delta 5), reused 68 (delta 1), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), done.
remote:
remote: Create a pull request for 'feature-update' on GitHub by visiting:
remote:   https://github.com/Sriarevalagan23/Test-repo-fork/pull/new/feature-
update
remote:
To https://github.com/Sriarevalagan23/Test-repo-fork
 * [new branch]      feature-update -> feature-update
srinidhi@Srinidhis-Laptop Test-repo-fork %

```

Step 6: Pull Latest Changes from Main Branch

Bash

git checkout main

git pull origin main

Screenshot 6: Pull latest changes.

```

To https://github.com/Sriarevalagan23/Test-repo-fork
 * [new branch]      feature-update -> feature-update
srinidhi@Srinidhis-Laptop Test-repo-fork % git checkout main

Switched to branch 'main'
Your branch is up to date with 'origin/main'.
[srinidhi@Srinidhis-Laptop Test-repo-fork % git pull origin main
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 930 bytes | 465.00 KiB/s, done.
From https://github.com/Sriarevalagan23/Test-repo-fork
 * branch                main                -> FETCH_HEAD
   12de6f2..d66e3ed      main                -> origin/main
Updating 12de6f2..d66e3ed
Fast-forward
 README.md | 3 +++
 1 file changed, 3 insertions(+)

```

Step 7: Switch Back to Feature Branch

Bash

git checkout feature-update

Step 8: Merge Main into Feature Branch

Bash

git merge main

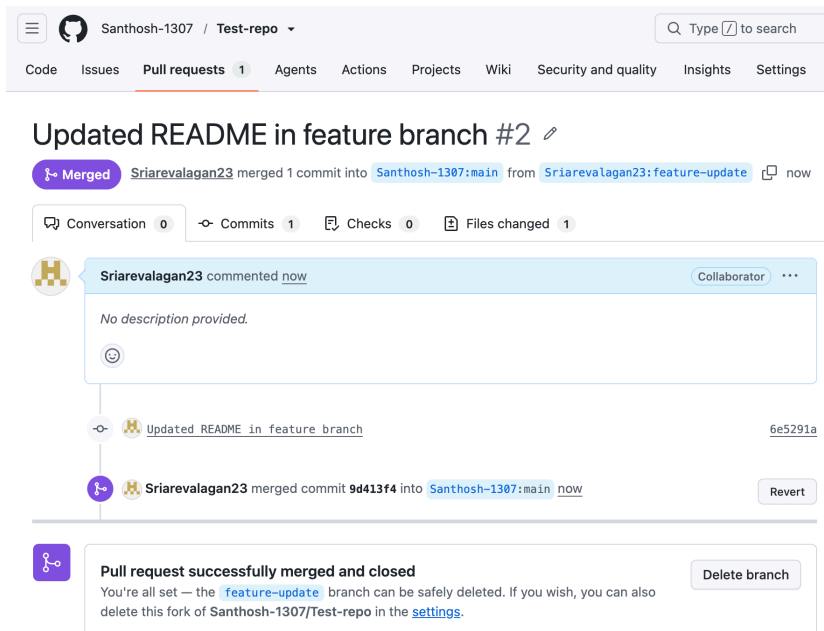
If no conflicts occur, the merge completes automatically.

If conflicts occur, Git displays a message similar to:

```

CONFLICT (content): Merge conflict in README.md
Automatic merge failed.

```



Screenshot 8: Merge

Step 9: Resolve Merge Conflict

Open the conflicted file.

Example:

```
<<<<<<< HEAD
Feature branch content
=====
Main branch content
>>>>>>> main
```

Edit the file manually to keep the required content.

Example after resolving:

```
Feature branch content
Main branch content
```

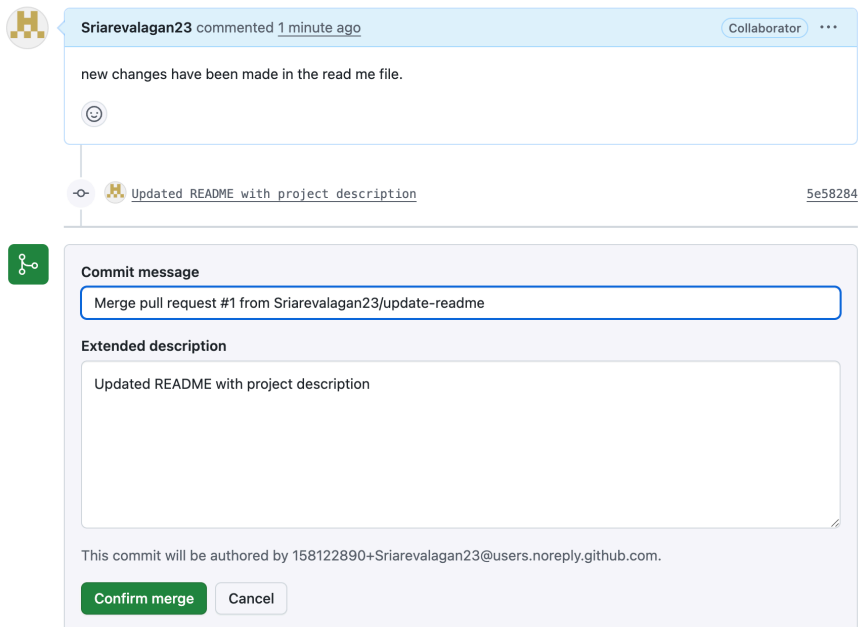
Mark the conflict as resolved:

```
Bash
git add README.md
```

Step 10: Commit the Merge

```
Bash
git commit -m "Resolved merge conflict"
```

Screenshot 10: Merge commit.



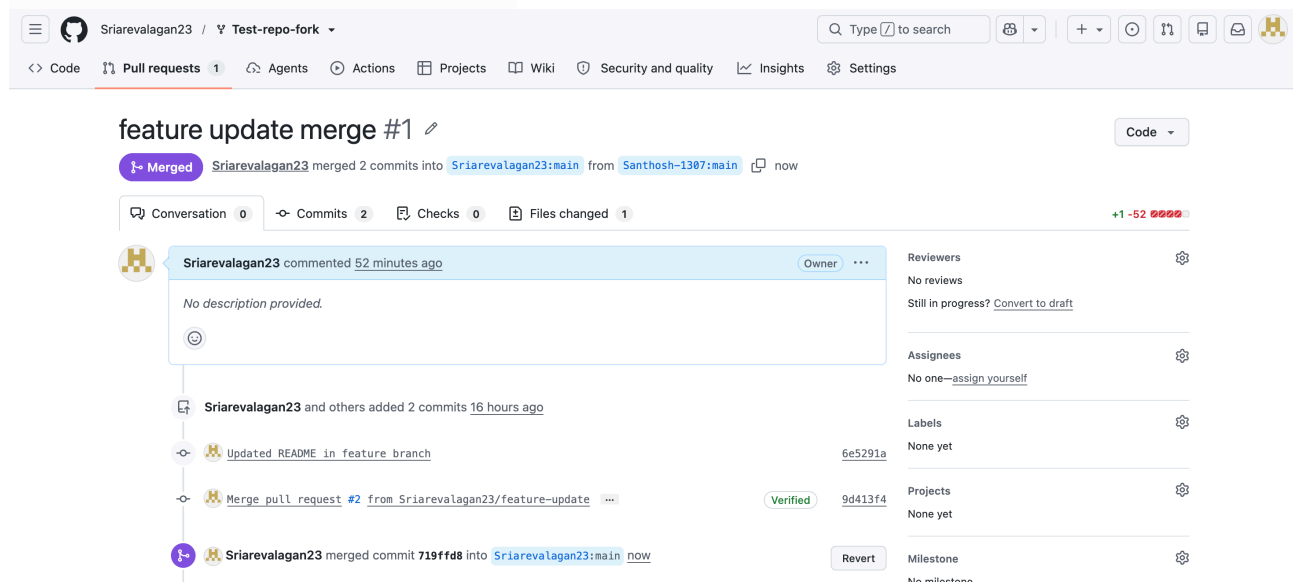
Step 11: Push the Updated Branch

Bash

```
git push origin feature-update
```

Go to GitHub and create a Pull Request.

Screenshot 11: Pull Request page.



Result

Git branching, merging, and manual merge conflict resolution were successfully performed. The updated feature branch was pushed to GitHub, and a Pull Request was created successfully.